

EECE 306 Lab 2 notebook. Coordinate Systems, Points and Vector Components

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Fill in every TODO, run `publish('Lab02_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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Setup

```
clear; close all;
rng(306);
```

Point roundtrips, 1000 random points

```
r = randn(1000,3);
esph = max(em.vec.mag(em.coord.sph2c(em.coord.c2sph(r)) - r));
ecyl = max(em.vec.mag(em.coord.cyl2c(em.coord.c2cyl(r)) - r));
fprintf('spherical roundtrip max error = %.3e\n', esph);
fprintf('cylindrical roundtrip max error = %.3e\n', ecyl);
```

```
spherical roundtrip max error = 2.812e-15
cylindrical roundtrip max error = 2.979e-15
```

Poles and the phi convention

```
disp('c2sph([0 0 1]) = '); disp(em.coord.c2sph([0 0 1]))
disp('c2sph([0 0 -1]) = '); disp(em.coord.c2sph([0 0 -1]))
q = [ 1 1 0; -1 1 0; -1 -1 0; 1 -1 0 ];
p = em.coord.c2cyl(q);
fprintf('phi in the four quadrants = %.4f %.4f %.4f %.4f rad\n', p(:,2));
fprintf('all inside [0, 2*pi) = %d\n', all(p(:,2) >= 0 & p(:,2) < 2*pi));
```

```
c2sph([0 0 1]) =
    1    0    0
```

```
c2sph([0 0 -1]) =
    1.0000    3.1416    0
```

```
phi in the four quadrants = 0.7854 2.3562 3.9270 5.4978 rad
all inside [0, 2*pi) = 1
```

Vector component roundtrips and the invariant

```
A = randn(1000,3);
As = em.coord.vecC2Sph(A, r);
ev = max(em.vec.mag(em.coord.vecSph2C(As, r) - A));
em_ = max(abs(em.vec.mag(As) - em.vec.mag(A)));
fprintf('vector roundtrip (spherical pair) max error = %.3e\n', ev);
fprintf('magnitude change under conversion max = %.3e\n', em_);
Ac = em.coord.vecC2Cyl(A, r);
```

```
ev2 = max(em.vec.mag(em.coord.vecCyl2C(Ac, r) - A));  
fprintf('vector roundtrip (cylindrical pair) max error = %.3e\n', ev2);
```

```
vector roundtrip (spherical pair) max error = 1.088e-15  
magnitude change under conversion max      = 8.882e-16  
vector roundtrip (cylindrical pair) max error = 6.280e-16
```

The two position demonstration

The same Cartesian vector $A = 2x + 3y$, expressed in spherical components at two different positions.

```
A0 = [2 3 0];  
disp('at (0, 5, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [0 5 0]))  
disp('at (4, 0, 0), (Ar, Atheta, Aphi) ='); disp(em.coord.vecC2Sph(A0, [4 0 0]))  
% The vector stayed at 2x +3x. The shperical coordinates changed because the spherical unit vectors are dependent on their position  
% A position argument is mandatory for component conversion as it determines the orientation of the spherical coordinates at the location.  
% Point conversion is meaningless as they are the same and not relative to the coordinate directions.
```

```
at (0, 5, 0), (Ar, Atheta, Aphi) =  
3.0000    0.0000   -2.0000
```

```
at (4, 0, 0), (Ar, Atheta, Aphi) =  
2.0000    0.0000    3.0000
```

Interpretation

the roundtrip tests catches a bug where the two functions are inconsistent with each other, which can happen from incorrect formulas,

```
%or coordinates not in the right space. It does not catch if both functions make the same mistake, as converting back and forth would  
%would still give the same point. The invariant that catches this error is the vector magnitude. It is reference free because  
%because it does not depend on any coordinate systems.
```

Problems encountered

Some vectors modules didn't run correctly the first time due to either incorrect variables used or typos.

Full test suite

```
runTests  
%{  
EECE 306 toolbox test suite  
-----  
  
c =  
  
2.9979e+08  
  
eta =  
  
376.7303  
  
test_lab01 checks complete  
PASS test_lab01          0.06 s  
PASS test_lab02          0.19 s  
-----  
2 passed, 0 failed, 2 total  
All tests passing.  
%}
```

```
EECE 306 toolbox test suite  
-----
```

C =

2.9979e+08

eta =

376.7303

test_lab01 checks complete

PASS test_lab01 0.07 s

PASS test_lab02 0.02 s

2 passed, 0 failed, 2 total

All tests passing.